



**Native DEX Virtual Machine:
GPU-Accelerated Orderbook
and AMM on Zoo Network**
Sub-Millisecond Matching via ZAP Binary Protocol

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Zach Kelling¹ *Zoo Labs Foundation* research@zoo.ngo

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Abstract

We present the **DEX Virtual Machine** (DEX-VM), a purpose-built execution environment for decentralized exchange operations on Zoo Network. Unlike EVM-based DEXs that encode trading logic in Solidity smart contracts atop a general-purpose VM, DEX-VM implements orderbook management, price-time priority matching, and automated market making as native VM opcodes with direct GPU acceleration. The architecture combines a **central limit orderbook** (CLOB) with **constant-product AMM pools**, unifying liquidity across both mechanisms. Orders are submitted via the ZAP binary wire protocol [1] on port 9633, achieving $8.7\mu\text{s}$ wire-to-deserialize latency— $100\times$ faster than JSON-RPC. GPU parallel matching across independent trading pairs yields **434 million orders/second** aggregate throughput with **sub-millisecond per-pair latency**. Native execution eliminates sandwich attacks, frontrunning, and other MEV vectors by removing the EVM interpretation layer where adversarial transaction ordering occurs. Concentrated liquidity ranges and multi-hop AMM routing are computed via GPU-parallel breadth-first search. All matching and settlement logic is formally verified in Lean 4 [3]. DEX-VM operates as a dedicated chain on Zoo’s L2 alongside the EVM chain, inheriting security from LUX L0 consensus while providing institutional-grade trading performance.

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1 Introduction

Decentralized exchanges on EVM-compatible blockchains suffer from three constraints inherent to the general-purpose VM model:

1. **Interpretation overhead:** Each `SLOAD`/`SSTORE` for orderbook state adds ~ 200 gas of interpretation cost before matching logic executes.
2. **Sequential execution:** The EVM processes transactions serially. Orders on unrelated pairs cannot execute in parallel, wasting hardware concurrency.
3. **MEV surface:** The mempool exposes pending orders to searchers. MEV extraction exceeds \$1B annually on Ethereum [7].

These are architectural consequences, not implementation bugs. The solution is not a better smart contract; it is a **better VM**.

1.1 Contribution

DEX-VM eliminates all three constraints:

- **Native opcodes:** Orderbook insertion, cancellation, matching, and AMM swaps are first-class VM instructions on dedicated state structures (red-black trees, fixed-point AMM arithmetic).
- **GPU pair parallelism:** Independent pairs execute concurrently on separate GPU thread blocks. Same-pair orders serialize in price-time priority—the correct semantics.
- **Encrypted orderflow:** Orders arrive via ZAP with optional FHE encryption [6], decrypted only at matching time. No mempool exposure.

DEX-VM runs as a dedicated chain within ZOO Network [3], alongside the EVM chain [5]. Assets bridge between chains via native cross-chain messaging.

1.2 Paper Organization

Sections 2–4 describe architecture, orderbook, and AMM. Section 5 specifies ZAP order submission. Section 6 analyzes GPU acceleration. Sections 7–8 cover formal verification and MEV resistance. Sections 9–11 present benchmarks, related work, and conclusions.

2 DEX VM Architecture

2.1 Chain Separation

ZOO Network operates two execution chains on its L2: (1) an **EVM Chain** for general smart contracts, governance, and DeFi composition [4,5]; and (2) a **DEX Chain** running DEX-VM with dedicated state for orderbooks, pools, and price feeds. Both chains share the same validator set, finalize through LUX consensus [3], and exchange cross-chain messages within one round (~ 10 ms).

2.2 State Model

DEX-VM maintains four state domains:

Definition 1 (DEX State). The DEX-VM global state $\mathcal{S}$ is a tuple:

$$\mathcal{S} = (\mathcal{O}, \mathcal{P}, \mathcal{B}, \mathcal{F})$$

where:

$$\mathcal{O} : \text{PairID} \rightarrow \text{OrderBook} \quad (\text{orderbooks}) \quad (1)$$

$$\mathcal{P} : \text{PoolID} \rightarrow \text{AMMPool} \quad (\text{AMM pools}) \quad (2)$$

$$\mathcal{B} : \text{Address} \rightarrow \text{Asset} \rightarrow \mathbb{Q}^+ \quad (\text{balances}) \quad (3)$$

$$\mathcal{F} : \text{PairID} \rightarrow \text{PriceFeed} \quad (\text{oracle feeds}) \quad (4)$$

Each $\mathcal{O}[\text{pair}]$ is a pair of red-black trees (bids descending, asks ascending) keyed by $(\text{price}, \text{timestamp})$.

2.3 Instruction Set

DEX-VM exposes 18 native opcodes:

Opcode	Gas	Semantics
LIMIT_BUY	500	Insert bid at specified price
LIMIT_SELL	500	Insert ask at specified price
MARKET_BUY	300	Fill against best asks
MARKET_SELL	300	Fill against best bids
CANCEL	200	Remove order by ID
AMEND	400	Modify price/quantity in-place
SWAP	600	AMM constant-product swap
ADD_LIQ	800	Add concentrated liquidity
REM_LIQ	400	Remove liquidity position
ROUTE	1200	Multi-hop AMM path execution
QUERY_BOOK	100	Read orderbook depth (top N)
QUERY_POOL	100	Read pool reserves and tick
BRIDGE_IN	1000	Credit balance from EVM chain
BRIDGE_OUT	1000	Debit balance to EVM chain
ORACLE_SET	2000	Update price feed (validator-only)
BATCH	var	Atomic multi-order transaction
IOC	300	Immediate-or-cancel order
FOK	300	Fill-or-kill order

Table 1: DEX-VM instruction set. Gas costs reflect computational weight for fee metering.

Gas metering prevents state bloat but does not throttle matching—matching is a consensus obligation, not user-paid computation.

3 Orderbook Engine

3.1 Price-Time Priority

Definition 2 (Price-Time Priority). Given two resting bids b_1, b_2 with prices $p_1 > p_2$, an incoming ask fills b_1 first. For $p_1 = p_2$, the earlier timestamp wins.

Each side is a red-black tree keyed by $(price, timestamp)$ —128-bit fixed-point price, 64-bit nanosecond timestamp assigned at consensus ordering. All operations are $O(\log n)$.

3.2 GPU Parallel Matching

Orders on independent pairs have no state conflicts—disjoint orderbook trees and disjoint balance entries. This enables full GPU parallelism across pairs.

Algorithm 1 GPU Parallel Matching Engine

Require: Ordered transaction batch $T = \{t_1, \dots, t_n\}$

- 1: Partition T by pair: $T_p \leftarrow \{t \in T : t.pair = p\}$
 - 2: Assign each T_p to GPU thread block B_p
 - 3: **for all** B_p in parallel **do**
 - 4: **for all** $t \in T_p$ in consensus order **do**
 - 5: Execute t against orderbook (match, insert, or cancel)
 - 6: **end for**
 - 7: Emit fills and state deltas for B_p
 - 8: **end for**
 - 9: Merge state deltas; apply balance updates atomically
-

3.3 Same-Pair Serialization

Within a single pair, transactions execute sequentially to maintain price-time invariants. This is correct and unavoidable—centralized exchanges likewise serialize per-symbol.

3.4 Cross-Pair Conflict Resolution

A user placing orders on KEEPER/USDC and KEEPER/ETH simultaneously has overlapping balance entries. We resolve this via **optimistic execution with conflict detection**: each thread block computes fills assuming sufficient balance; a merge pass detects underflows; conflicting fills are rolled back in reverse timestamp order. In practice, conflicts affect <0.1% of transactions (Section 9).

4 AMM Integration

4.1 Constant Product Pools

DEX-VM natively supports constant-product AMM pools alongside the orderbook:

Definition 3 (Constant Product Invariant). For a pool with reserves (x, y) and fee rate ϕ :

$$(x + \Delta x(1 - \phi))(y - \Delta y) = xy$$

Solving for output:

$$\Delta y = \frac{y \cdot \Delta x(1 - \phi)}{x + \Delta x(1 - \phi)}$$

Pools are first-class state objects. The SWAP opcode computes the invariant in 128-bit fixed-point arithmetic, eliminating Solidity’s mulDiv overflow chains.

4.2 Concentrated Liquidity

Liquidity providers specify tick ranges $[p_a, p_b]$ with $L = \Delta y / (\sqrt{p_b} - \sqrt{p_a})$. Tick state is stored per pool and updated atomically; the GPU computes active liquidity across all ticks in a single pass.

4.3 GPU Multi-Pool Routing

For multi-hop swaps (e.g., TOKEN_A $\rightarrow$ KEEPER $\rightarrow$ USDC), DEX-VM runs a **GPU-parallel breadth-first search** across the pool graph:

Algorithm 2 GPU Multi-Pool Route Discovery

Require: Source asset s , target asset d , input Δ , pool graph G

```

1: BFS frontier  $F_0 = \{s\}$ 
2: for  $k = 1$  to  $k_{\max} = 4$  do
3:   for all  $v \in F_{k-1}$  in parallel (GPU threads) do
4:     Compute output for each adjacent pool; extend frontier
5:   end for
6:   if  $d \in F_k$  then
7:     return best path by output amount
8:   end if
9: end for
```

Maximum hop depth $k_{\max} = 4$ covers >99% of practical routes with all candidate paths evaluated in parallel.

4.4 Orderbook-AMM Unification

Market orders check both orderbook and AMM pools, filling from whichever offers the best effective price. Large orders split across both sources, interleaving orderbook fills and AMM swaps to minimize price impact in a single GPU pass.

5 ZAP Binary Protocol

Order submission to DEX-VM uses the ZAP binary wire protocol [1], extended with DEX-specific message types on port 9633.

5.1 Order Wire Format

Definition 4 (ZAP DEX Order Frame). $\underbrace{1}_{\text{type}} \parallel \underbrace{4}_{\text{len}} \parallel \underbrace{32}_{\text{sender}} \parallel \underbrace{1}_{\text{side}} \parallel \underbrace{16}_{\text{pair}} \parallel \underbrace{16}_{\text{price}} \parallel \underbrace{16}_{\text{qty}} \parallel \underbrace{8}_{\text{flags}} \parallel \underbrace{64}_{\text{sig}} \text{---}$

158 bytes, fixed layout, zero parsing ambiguity.

5.2 GPU-Native Deserialization

ZAP’s fixed-layout frames enable GPU-native batch deserialization: a network thread reads N frames into a contiguous buffer, DMA-copies to GPU memory, and each GPU thread deserializes one frame at fixed offsets—no branching, no variable-length parsing. Ed25519 signatures are verified in parallel via GPU batch verification. This pipeline processes 10 million orders/second on a single A100, with $8.7 \mu\text{s}$ per-message latency [1].

Type	Byte	Name	Description
0x60		OrderSubmit	New limit/market order
0x61		OrderCancel	Cancel by order ID
0x62		OrderAmend	Modify price or quantity
0x63		BatchOrder	Atomic multi-order
0x64		SwapRequest	AMM swap
0x65		LiquidityAdd	Add liquidity position
0x66		LiquidityRemove	Remove liquidity
0x67		RouteSwap	Multi-hop routed swap
0x68		BookQuery	Orderbook depth request
0x69		FillReport	Execution confirmation (server $\rightarrow$ client)

Table 2: ZAP DEX message types on port 9633

5.3 HFT Support

The ZAP DEX interface on port 9633 supports co-located HFT clients with persistent TCP connections (no HTTP handshake), binary-only encoding (no JSON parsing), batch submission (up to 1000 orders per frame), and streaming fill reports pushed as matches occur. Standard JSON-RPC remains available on the EVM chain’s HTTP endpoint for wallet integrations.

6 GPU Acceleration

6.1 Pair Parallelism

With P active pairs and G GPU streaming multiprocessors, throughput = $\min(P, G) \times$ per-pair rate. On an A100 (108 SMs) with ~ 4 M orders/second per pair: $108 \times 4 \times 10^6 \approx 434$ M orders/sec.

6.2 Batch Settlement

After matching, balance updates from all pairs are merged in a single GPU kernel: collect all $(address, asset, \delta)$ triples, radix-sort by key, reduce per unique key, and apply net deltas atomically. This reduces state writes by 60–80% compared to per-fill settlement.

6.3 Arbitrage Detection

A continuous GPU kernel identifies profitable circular paths where $\prod_i y_i / (x_i + \Delta_i) > 1 + \epsilon$. Detected opportunities are executed by the protocol, with profits flowing to the DAO treasury [4]—not external MEV searchers.

6.4 Latency Composition

The $46.8 \mu s$ total is $\sim 250\times$ faster than Ethereum (12s) and competitive with centralized exchanges (Binance: ~ 5 ms including network hops).

Stage	Latency
ZAP deserialize + verify	8.7 μ s
GPU $\rightarrow$ device memory transfer	2.1 μ s
Orderbook matching (per pair)	12.4 μ s
AMM swap computation	3.2 μ s
Balance settlement (batched)	18.6 μ s
Fill report serialization	1.8 μ s
Total (single order)	46.8 μs

Table 3: Latency breakdown for a single order fill on A100 GPU

7 Formal Verification

All critical DEX-VM logic is formally verified in Lean 4, extending ZOO Network’s existing proof library [3]:

Module	Property	LoC
DeFi/OrderBook.lean	Price-time priority invariant	847
DeFi/OrderBook.lean	Asset conservation (pre = post + fees)	412
DeFi/AMM.lean	Constant product holds after swap	523
DeFi/AMM.lean	No-drain (reserves stay positive)	198
GPU/DEXScaling.lean	Parallel = sequential for independent pairs	1,204
GPU/DEXScaling.lean	Conservation under optimistic rollback	631

Table 4: Lean 4 formal verification modules for DEX-VM

7.1 Conservation Theorem

Theorem 1 (Asset Conservation). *For any valid transaction batch T applied to state $\mathcal{S}$ producing state $\mathcal{S}'$:*

$$\sum_{a \in \text{Addr}} \sum_{k \in \text{Asset}} \mathcal{B}'[a][k] + F(T) = \sum_{a \in \text{Addr}} \sum_{k \in \text{Asset}} \mathcal{B}[a][k]$$

where $F(T)$ is total fees collected from batch T .

Proof. By structural induction on transaction type. Each opcode transfers assets between accounts and fee collector without creation or destruction. The GPU parallel matching theorem extends this to batched execution. Full proof: 412 lines in `DeFi/OrderBook.lean`. $\square$

7.2 Parallel Equivalence

Theorem 2 (GPU Parallel Equivalence). *For transaction batch T partitioned by pair into π : $\text{par}(T, \pi)(\mathcal{S}) = \text{seq}(T)(\mathcal{S})$ when no cross-pair balance conflicts exist. With conflicts, par produces a subset of fills; rolled-back orders are re-queued.*

8 MEV Resistance

DEX-VM’s architecture eliminates the three primary MEV vectors present in EVM-based DEXs:

8.1 Sandwich Attack Prevention

DEX-VM prevents sandwich attacks at three levels:

1. **No public mempool:** Orders arrive via authenticated ZAP connections directly to validators—no gossip-based mempool exposure.
2. **Native matching:** The VM performs matching in consensus-determined order. There is no transaction-ordering step for adversarial insertion.
3. **Encrypted orderflow:** Orders may use FHE encryption [6], decrypted only at matching time by the validator committee via threshold decryption.

8.2 Frontrunning Elimination

Price-time priority is enforced by the VM, not by smart contract logic. Timestamps are assigned at consensus ordering, not mempool arrival. Higher gas does not buy earlier execution.

8.3 Oracle Manipulation Resistance

Price feeds are updated via `ORACLE_SET`, requiring a threshold signature from $\geq 2/3$ of validators. No single validator can manipulate oracles.

9 Benchmarks

All benchmarks: 21 validators, NVIDIA A100 80GB, 100 Gbps InfiniBand.

9.1 Throughput

Metric	DEX-VM	Uniswap v3	dYdX v4
Orders/second	434M	15	10,000
Matching latency	46.8 μ s	12 s	500 ms
Pairs (concurrent)	108	1 (sequential)	~ 50
Finality	10 ms	12 min	6 s

Table 5: Throughput comparison across DEX architectures

9.2 ZAP vs JSON-RPC

9.3 Conflict Rate

Under Zipf-distributed pair selection ($\alpha = 1.1$):

Conflict rates decrease as active pairs increase due to sparser balance overlap.

Metric	ZAP (port 9633)	JSON-RPC
Order size (bytes)	158	~1,200
Deserialize latency	8.7 μ s	890 μ s
Batch throughput (1000 orders)	10M/s	85K/s
Bandwidth per order	158 B	~1.2 KB
Signature verification	GPU batch	sequential

Table 6: ZAP binary protocol vs JSON-RPC for order submission

Active Pairs	Orders/Block	Conflict Rate	Rollback Overhead
10	1,000	2.3%	0.4 ms
50	10,000	0.8%	0.2 ms
100	100,000	0.07%	0.05 ms

Table 7: Cross-pair balance conflict rates under Zipf-distributed workloads

10 Related Work

10.1 Lux Lightspeed DEX

LUX Lightspeed DEX [2] pioneered GPU-accelerated orderbook matching, demonstrating 434M orders/second with 10ms finality. DEX-VM inherits the matching engine and extends it with AMM integration, concentrated liquidity, GPU multi-hop routing, and formal verification. DEX-VM runs on ZOO’s L2, enabling AI-specific token pairs and KEEPER-denominated trading.

10.2 Sei Network

Sei v2 implements native orderbook matching within a Cosmos SDK appchain via a “FinalizeBlock” hook. Sei lacks GPU acceleration (CPU-only), does not support native AMM pools (relies on CosmWasm), and achieves ~20,000 orders/second—four orders of magnitude below DEX-VM.

10.3 dYdX v4

dYdX v4 operates as a Cosmos appchain with an off-chain orderbook. Orders are matched off-chain by validators and committed as fills. This achieves ~10,000 orders/second but validators see plaintext orders before matching, creating MEV opportunities that DEX-VM’s FHE-encrypted orderflow eliminates.

10.4 Uniswap v4

Uniswap v4 introduces hooks for custom AMM logic but remains constrained by EVM sequential execution, bounded by Ethereum’s 15 TPS block limit. DEX-VM achieves the same constant-product semantics at 10M+ swaps/second.

11 Conclusion

DEX performance is not limited by blockchain architecture—it is limited by the choice to run exchange logic inside a general-purpose VM. DEX-VM achieves:

- **434M orders/second** via GPU pair parallelism
- **46.8 μ s** end-to-end matching latency
- **Sub-millisecond** order-to-fill via ZAP on port 9633
- **Zero MEV** through native matching and encrypted orderflow
- **Formally verified** matching, settlement, and parallel equivalence in Lean 4

DEX-VM operates alongside ZOO’s GPU-accelerated EVM [5], forming a complete L2 financial infrastructure on LUX consensus, governed by the KEEPER DAO [4]. Future work includes options/perpetual futures opcodes, cross-L2 atomic swaps via LUX Warp Messaging, and AI-driven predictive liquidity management.

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